

Amazon Product Performance Analysis Using SQL

Abstract

This project will use SQL to analyze Amazon product listing data to find how well products are performing, customer satisfaction trends over time, how well individual manufacturers are performing, pricing patterns and stock availability.

The purpose of this project is to show how SQL can assist in conducting business analysis and making management decisions by providing real-world e-commerce data.

Problem Statement

Amazon has an extensive range of products from numerous manufacturers, therefore, understanding customer preferences, product quality, pricing strategies, and inventory levels will help improve performance.

This project aims to answer the following questions:

- What products are rated top rated?
- Which manufacturer has the best performance?
- How does pricing impact product rating?
- What percentage of products are currently in stock?
- What products are in need of product quality improvement?
- How are products distributed among various rating categories?

Dataset Information

Dataset Name: Amazon Product Listing Dataset

Data Source: Publicly available Amazon marketplace product data

Type of Data: Secondary Data

Dataset Attributes

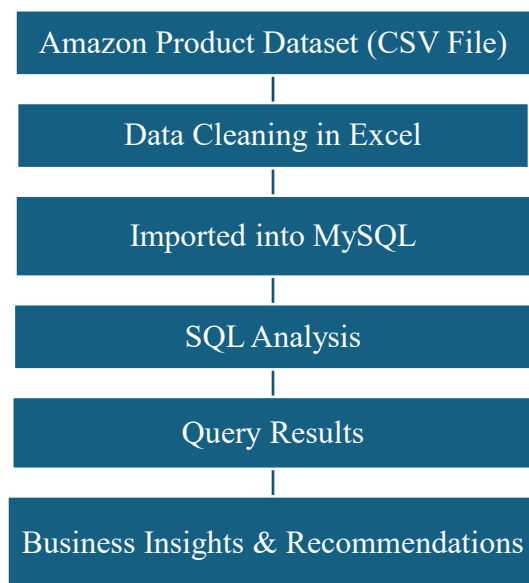
- Uniq Id
- Title
- Manufacturer
- Price
- Stock
- Average Rating

- Number Of Ratings
- Num Of Review

Tools Used

- MySQL Workbench
- SQL
- Microsoft Excel (Data Cleaning)

System Flow Diagram



Database Structure

Table Name:

amazon_project_data

Main Columns:

Uniq Id
Title
Manufacturer
Price
Stock
Average Rating

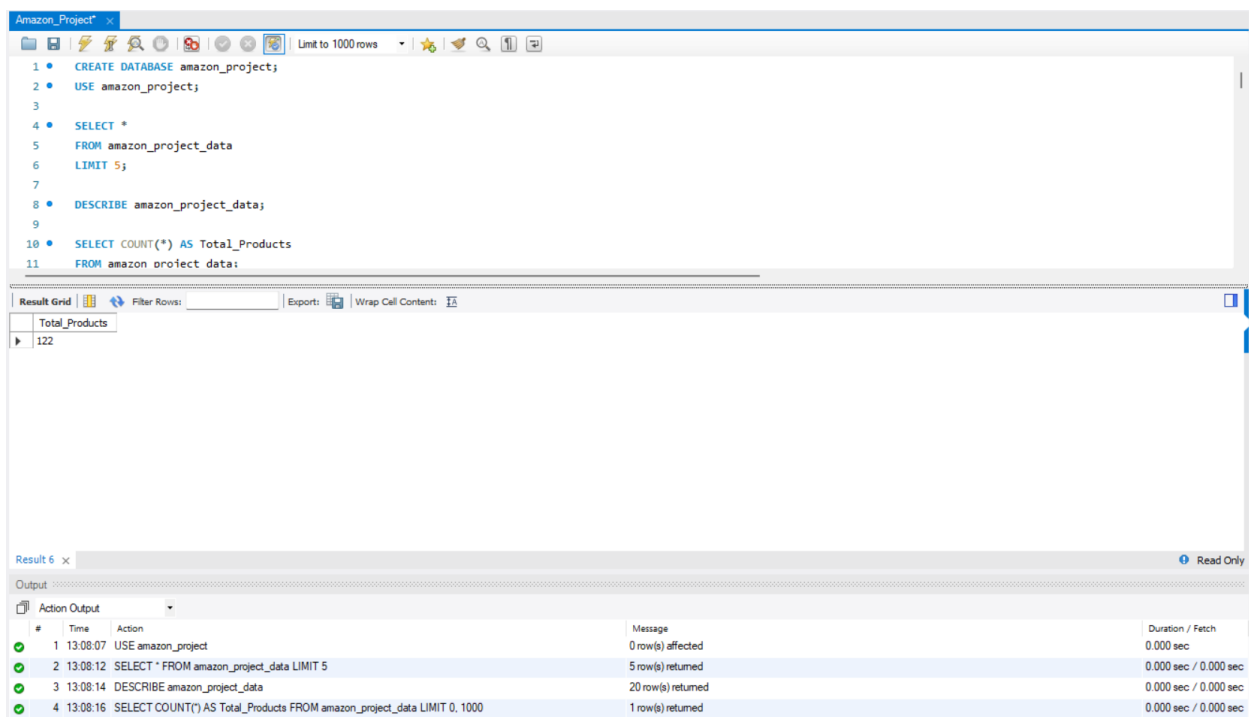
Number Of Ratings
Num Of Reviews

SQL Queries Performed

1) Total Number of Products: Determined the total number of products available in the dataset.

```
SELECT COUNT(*) AS Total_Products  
FROM amazon_project_data;
```

2) Top Rated Products: Identified products with the highest customer ratings.



The screenshot shows a SQL IDE window titled "Amazon_Project". The query editor contains the following SQL code:

```
1 • CREATE DATABASE amazon_project;  
2 • USE amazon_project;  
3  
4 • SELECT *  
5 FROM amazon_project_data  
6 LIMIT 5;  
7  
8 • DESCRIBE amazon_project_data;  
9  
10 • SELECT COUNT(*) AS Total_Products  
11 FROM amazon project data;
```

The "Result Grid" shows the results of the query. The first row is labeled "Total_Products" and contains the value "122".

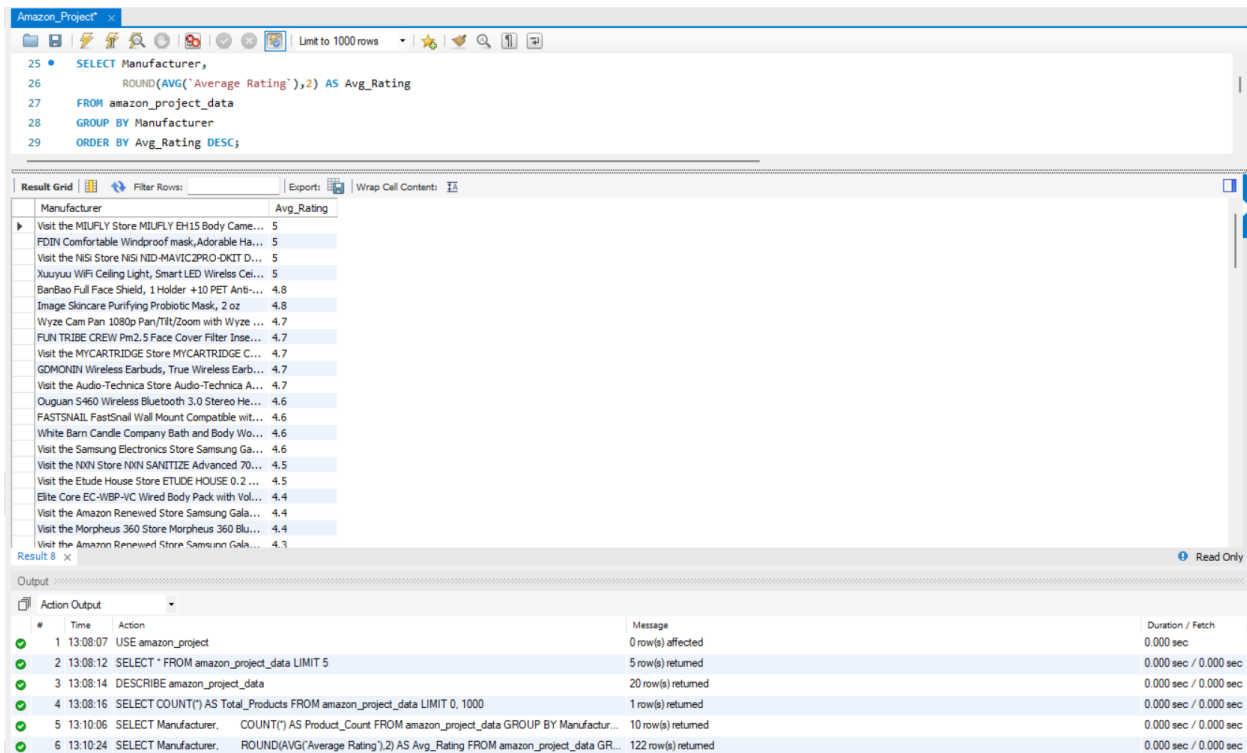
The "Output" pane shows the execution log:

#	Time	Action	Message	Duration / Fetch
1	13:08:07	USE amazon_project	0 row(s) affected	0.000 sec
2	13:08:12	SELECT * FROM amazon_project_data LIMIT 5	5 row(s) returned	0.000 sec / 0.000 sec
3	13:08:14	DESCRIBE amazon_project_data	20 row(s) returned	0.000 sec / 0.000 sec
4	13:08:16	SELECT COUNT(*) AS Total_Products FROM amazon_project_data LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

3) Top Manufacturers by Product Count: Identified manufacturers with the largest number of listed products.

```
SELECT Manufacturer,  
COUNT(*) AS Product_Count  
FROM amazon_project_data  
GROUP BY Manufacturer  
ORDER BY Product_Count DESC  
LIMIT 10;
```

- 4) **Average Rating by Manufacturer:** Evaluated overall manufacturer performance based on customer ratings.



```
25 SELECT Manufacturer,
26 ROUND(AVG('Average Rating'),2) AS Avg_Rating
27 FROM amazon_project_data
28 GROUP BY Manufacturer
29 ORDER BY Avg_Rating DESC;
```

Manufacturer	Avg_Rating
Visit the MJFLY Store MJFLY EH15 Body Came...	5
FDIN Comfortable Windproof mask, Adorable Ha...	5
Visit the NSI Store NSI NID-MAVIC2PRO-DKIT D...	5
Xuuyuu WiFi Ceiling Light, Smart LED Wireless Cel...	5
BarBao Full Face Shield, 1 Holder +10 PET Anti...	4.8
Image Skincare Purifying Probiotic Mask, 2 oz	4.8
Wyze Cam Pan 1080p Pan/Tilt/Zoom with Wyze ...	4.7
FUN TRIBE CREW Pm2.5 Face Cover Filter Inse...	4.7
Visit the MYCARTRIDGE Store MYCARTRIDGE C...	4.7
GDMONIN Wireless Earbuds, True Wireless Earb...	4.7
Visit the Audio-Technica Store Audio-Technica A...	4.7
Ouguan S460 Wireless Bluetooth 3.0 Stereo He...	4.6
FASTSNAIL FastSnail Wall Mount Compatible wit...	4.6
White Barn Candle Company Bath and Body Wo...	4.6
Visit the Samsung Electronics Store Samsung Ga...	4.6
Visit the NNN Store NNN SANITIZE Advanced 70...	4.5
Visit the Etude House Store ETUDE HOUSE 0.2 ...	4.5
Elite Core EC-WBP-VC Wired Body Pack with Vol...	4.4
Visit the Amazon Renewed Store Samsung Gala...	4.4
Visit the Morpheus 360 Store Morpheus 360 Blu...	4.4
Visit the Amazon Renewed Store Same inn Gala...	4.3

Result 8 x

Output

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4	13:08:16	SELECT COUNT(*) AS Total_Products FROM amazon_project_data LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
5	13:10:06	SELECT Manufacturer, COUNT(*) AS Product_Count FROM amazon_project_data GROUP BY Manufactur...	10 row(s) returned	0.000 sec / 0.000 sec
6	13:10:24	SELECT Manufacturer, ROUND(AVG('Average Rating'),2) AS Avg_Rating FROM amazon_project_data GR...	122 row(s) returned	0.000 sec / 0.000 sec

- 5) **Poorly Rated Products:** Identified products requiring quality improvement.

- 6) **Products In Stock:** Measured product availability.

```
SELECT COUNT(*) AS Products_In_Stock
FROM amazon_project_data
WHERE Stock = 'TRUE';
```

- 7) **Products Out Of Stock:** Identified unavailable products.

```
SELECT COUNT(*) AS Products_Out_Of_Stock
FROM amazon_project_data
WHERE Stock = 'FALSE';
```

- 8) **Customer Satisfaction Analysis:** Classified products into satisfaction categories.

Categories:

- Satisfied
- Neutral
- Dissatisfied

The screenshot shows a SQL IDE window titled "Amazon_Project". The query editor contains the following SQL code:

```

91
92 SELECT
93 CASE
94     WHEN 'Average Rating' >= 4 THEN 'Satisfied'
95     WHEN 'Average Rating' >= 3 THEN 'Neutral'
96     ELSE 'Dissatisfied'
97 END AS Customer_Satisfaction,
98 COUNT(*) AS Products
99 FROM amazon_project_data
100 GROUP BY Customer_Satisfaction;
101

```

The "Result Grid" shows the following data:

Customer_Satisfaction	Products
Satisfied	35
Dissatisfied	73
Neutral	14

The "Output" pane shows the execution log:

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6	13:10:24	SELECT Manufacturer, ROUND(AVG(Average Rating),2) AS Avg_Rating FROM amazon_project_data G...	122 row(s) returned	0.000 sec / 0.000 sec
7	13:11:29	SELECT CASE WHEN 'Average Rating' >= 4 THEN 'Satisfied' WHEN 'Average Rating' >= 3 THEN Neu...	3 row(s) returned	0.000 sec / 0.000 sec

9) Rating Distribution Analysis: Analyzed overall product quality distribution.

10) Top Products by Number of Ratings: Identified the most popular products.

11) High Price but Low Rating Products: Identified overpriced products that may not satisfy customers.

```

SELECT Title,
        Price,
        `Average Rating`
FROM amazon_project_data
WHERE Price <> 'NA'
AND CAST(Price AS DECIMAL(10,2)) > 100
AND `Average Rating` < 3;

```

12) Price Category Analysis: Compared customer ratings across different price segments.

Categories:

- Budget
- Mid Range
- Premium

```

130 SELECT
131 CASE
132     WHEN CAST(Price AS DECIMAL(10,2)) < 50 THEN 'Budget'
133     WHEN CAST(Price AS DECIMAL(10,2)) BETWEEN 50 AND 200 THEN 'Mid Range'
134     ELSE 'Premium'
135 END AS Price_Category,
136 COUNT(*) AS Products,
137 ROUND(AVG('Average Rating'),2) AS Avg_Rating
138 FROM amazon_project_data
139 WHERE Price <> 'NA'
140 GROUP BY Price_Category;
141

```

Price_Category	Products	Avg_Rating
Budget	42	2.05
Mid Range	14	1.96
Premium	7	2.61

#	Time	Action	Message	Duration / Fetch
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7	13:11:29	SELECT CASE WHEN 'Average Rating' >= 4 THEN 'Satisfied' WHEN 'Average Rating' >= 3 THEN 'Neu...	3 row(s) returned	0.000 sec / 0.000 sec
8	13:12:25	SELECT CASE WHEN CAST(Price AS DECIMAL(10,2)) < 50 THEN 'Budget' WHEN CAST(Price AS DEC...	3 row(s) returned	0.015 sec / 0.000 sec

Business Insights

- **Product Performance-** Higher rated products typically signify a greater degree of customer satisfaction, as well as a higher quality product.
- **Manufacturer Performance-** A manufacturer analysis can help show which manufacturers consistently produce good quality products.
- **Customer Satisfaction -** Customer ratings can be used as an indicator of performance and assist in evaluating a product's quality.
- **Inventory Analysis-** Inventory Availability Analysis provides information about the accessibility of the products being evaluated, and how they are being managed in terms of supply.
- **Pricing Strategy-** An analysis of the category of price will help determine if higher priced products lead to higher customer satisfaction.
- **Product Improvement-** Products with poor ratings identify opportunities for the manufacturer to improve overall quality.

Key Findings

- A few products had high customer ratings showing strong product quality.
- Retailer performance significantly differs from product to product.

- Retailers show varying stock levels between retailers.
- A small number of expensive products had low customer ratings meaning that just because something costs more than average does not mean that customers will be satisfied with the purchase.
- Ratings are clustered in certain areas of the ranking system.
- Customer satisfaction can be assessed effectively via rating analysis.

Recommendations

- 1) Increase the level of customer interaction by promoting only positively rated products.
- 2) Conduct assessments of poorly rated products in terms of overall quality and investigate specific problems with them.
- 3) Build and nurture relationships with manufacturers that consistently produce high-quality products.
- 4) Examine current practices for managing inventory for out-of-stock items.
- 5) Examine and adjust pricing strategies for high-cost and poorly rated products.
- 6) Use customer ratings to evaluate products as an important measure of performance.

Learning Outcomes

Through this project, the following SQL concepts were applied:

- SELECT
- WHERE
- ORDER BY
- GROUP BY
- HAVING
- COUNT()
- AVG()
- ROUND()
- CASE Statements
- Aggregate Functions
- Data Analysis Queries

Conclusion

By successfully applying SQL to provide insight into Amazon products, this project generated valuable information regarding product performance, how satisfied customers are with those products and whether or not the manufacturer has produced enough of those products so they will not be out of stock. Additionally, this analysis demonstrates how powerful an application like SQL can be in order to provide insightful solutions to real business problems. These insights could prove to be quite useful for an organization looking to improve product quality and its customers' experience with those products; however, they can also help improve overall business performance.